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Title: A Comparative Study of Ensemble Learning Techniques for Customer Churn Prediction with Decision Threshold Optimization

1. Introduction

In today's highly competitive business environment, retaining customers has become just as important if not more important than acquiring new ones. Companies invest significant resources into marketing and customer acquisition, yet even a small increase in customer retention can lead to a substantial increase in profitability. This is especially true in industries such as telecommunications, where customers can easily switch to competitors offering similar services. Customer churn refers to the phenomenon where a customer stops using a company's service. Understanding and predicting churn behavior allows organizations to take proactive steps to retain customers before they leave. Traditional approaches to churn analysis often relied on manual rules or simple statistical techniques, which are limited in their ability to capture complex patterns in large datasets.

With the advancement of machine learning, it has become possible to analyze customer data more effectively and uncover hidden patterns that indicate potential churn. However, individual machine learning models often face challenges such as overfitting, sensitivity to noise, and difficulty handling imbalanced datasets. These limitations can reduce their effectiveness in real-world scenarios.

To overcome these challenges, ensemble learning techniques have gained significant attention. Ensemble methods combine multiple models to produce more accurate and robust predictions. By leveraging the strengths of different algorithms, ensemble learning reduces errors and improves generalization.

This report presents a detailed study on customer churn prediction using various ensemble learning techniques. It also explores the impact of threshold optimization, which is often overlooked but plays a crucial role in improving performance in imbalanced classification problems.

2. Objectives

The goal of this study is not only to build predictive models but also to understand their behavior and limitations in a real-world context.
The key objectives of this report are:

- To develop and evaluate machine learning models for predicting customer churn
- To compare the performance of individual models with ensemble techniques
- To analyze the effect of class imbalance on prediction performance
- To investigate how ensemble learning improves robustness and accuracy
- To apply threshold optimization to enhance classification results
- To identify important factors that influence customer churn

Beyond achieving high accuracy, this study focuses on building models that are practically useful, especially in identifying customers who are likely to churn.

3. Dataset Description

The dataset used in this study is an updated version of the IBM Telco Customer Churn dataset (version 11.1.3+). This dataset is widely used in academic research and provides a realistic representation of customer behavior in the telecommunications industry.

3.1 Overview of the Dataset

The dataset contains detailed information about customers, including their demographics, services, and billing information. Each row represents a single customer, and each column represents a specific attribute.

The dataset includes:

- Customer demographic information
- Subscription and service usage details
- Billing and payment information
- Customer tenure and contract type

3.2 Feature Categories

To better understand the dataset, the features can be grouped into categories:

a. Demographic Features: These include attributes such as gender, senior citizen status, and whether the customer has dependents. These features help in understanding customer profiles.
b. Account Information: This includes tenure, contract type, and billing preferences. These features often have a strong relationship with churn behavior.
c. Service Features: These indicate whether the customer uses services such as internet, streaming, security, and support.
d. Financial Features: Monthly charges and total charges provide insights into the customer's spending behavior.

3.3 Target Variable

The target variable is:

- **Churn Value**
  - 1 → Customer has churned
  - 0 → Customer is retained

This is a binary classification problem.

3.4 Class Imbalance

One of the key challenges in this dataset is class imbalance. The number of customers who churn is significantly lower than those who do not. This imbalance can cause machine learning models to become biased toward the majority class, leading to misleadingly high accuracy but poor detection of churn cases.

4. Data Preprocessing

Before applying machine learning models, the dataset must be cleaned and transformed into a suitable format.

4.1 Data Cleaning

The dataset contained some inconsistencies, particularly in the **Total Charges** column, which was stored as a non-numeric value. This was converted into a numeric format.

Missing values were handled using **median imputation**, which is robust to outliers and ensures that the dataset remains stable.

4.2 Feature Transformation

Categorical variables cannot be directly used by most machine learning algorithms. Therefore, **one-hot encoding** was applied to convert categorical variables into numerical form.

4.3 Feature Selection and Leakage Prevention

A critical step in preprocessing was removing features that could lead to **data leakage**. Data leakage occurs when information that would not be available at prediction time is included in the model, leading to overly optimistic results.

The following features were removed:

- Customer ID
- Location-based attributes
- Churn-related indicators such as churn score and churn reason

4.4 Train-Test Split

The dataset was split into training and testing sets using an 80:20 ratio. Stratified sampling was used to ensure that both sets maintain the same class distribution.

5. Methodology

This study follows a structured and practical approach to building a customer churn prediction system. Instead of relying on a single model, multiple machine learning techniques are explored and combined to improve performance. The overall process involves preparing the data, training individual models, combining them using ensemble techniques, and finally optimizing the decision-making process through threshold tuning.

5.1 Data Preparation and Feature Handling

Before any model training begins, the dataset is carefully prepared to ensure that the models receive clean and meaningful input.

One important step in this process is converting all features into a numerical format. Since machine learning algorithms cannot directly process categorical values such as “Yes” or “No,” these features are transformed using one-hot encoding. This creates new binary columns representing each category, allowing models to interpret them correctly.

Another critical aspect is handling missing and inconsistent data. For example, the “Total Charges” column is converted into a numeric format, and any missing values are filled using the median. This approach ensures that extreme values do not distort the dataset while maintaining overall data integrity.

Additionally, special attention is given to removing features that could introduce **data leakage**. Features such as customer ID or churn-related labels (e.g., churn score or churn reason) are excluded because they either do not contribute meaningful predictive information or contain information that would not be available in real-world prediction scenarios.

Once the data is cleaned and transformed, it is divided into training and testing sets. Stratified sampling is used to maintain the original class distribution, which is particularly important due to the imbalance between churn and non-churn cases.

5.2 Training of Base Models

The next step involves training several individual machine learning models, referred to as base models. Each model learns patterns from the training data in a different way.

- **Logistic Regression** learns a linear relationship between features and the probability of churn. It serves as a simple and interpretable baseline model.
- **Random Forest** builds multiple decision trees using random subsets of the data and features. Each tree makes a prediction, and the final output is determined by aggregating these predictions. This reduces overfitting and improves generalization.
- **XGBoost** is a boosting algorithm that builds models sequentially. Each new model focuses on correcting the mistakes made by previous ones, making it highly effective for structured data.
- **Multi-Layer Perceptron (MLP)** is a neural network that captures complex nonlinear relationships between features and the target variable.

During training, each model learns patterns such as how customer tenure, service type, or billing behavior relate to churn. However, these models may also memorize patterns too closely, leading to overfitting, which is later analyzed.

5.3 Ensemble Learning Approach

After training the base models, ensemble techniques are applied to combine their predictions. The idea behind ensemble learning is that multiple models working together can produce better results than any single model alone.

Several ensemble methods are implemented:

Voting Classifier

In voting, predictions from multiple models are combined:

- In **hard voting**, each model casts a vote for a class, and the majority wins.
- In **soft voting**, probabilities from each model are averaged, and the final prediction is based on the highest probability.

This approach is simple but effective when models have diverse strengths.

Bagging (Bootstrap Aggregating)

Bagging works by training multiple versions of the same model (in this case, decision trees) on different subsets of the training data. These subsets are created through random sampling with replacement.

Each model learns slightly different patterns, and their predictions are averaged. This reduces variance and helps prevent overfitting. In this study, bagging proves to be one of the most effective techniques.

Stacking

Stacking takes a more advanced approach by combining multiple base models using a meta-model. Instead of directly averaging predictions, the outputs of base models are used as inputs for another model (typically logistic regression).

This meta-model learns how to best combine the predictions of base models, effectively capturing relationships between their outputs. As a result, stacking can often achieve strong performance.

Boosting (AdaBoost)
Boosting focuses on improving model performance by giving more importance to misclassified samples. In each iteration, the model pays more attention to difficult cases, attempting to correct previous errors.
However, this approach can be sensitive to noise and class imbalance, which may negatively affect performance in churn prediction tasks.

5.4 Model Evaluation
Once all models are trained, their performance is evaluated using multiple metrics.
Accuracy alone is not sufficient because the dataset is imbalanced. For example, a model could achieve high accuracy by simply predicting that no customers will churn. Therefore, additional metrics are used:

- **Precision** measures how many predicted churn cases are actually correct
- **Recall** measures how many actual churn cases are correctly identified
- **F1-score** provides a balance between precision and recall
- **ROC-AUC** evaluates the model's ability to distinguish between classes across different thresholds

By analyzing these metrics together, a more complete understanding of model performance is obtained.

5.5 Threshold Optimization
Most classification models use a default threshold of 0.5 to decide whether a customer will churn. However, this threshold is not always optimal, especially in imbalanced datasets.
In this study, a range of threshold values is tested. For each threshold, predictions are generated, and the corresponding F1-score is calculated. The threshold that produces the highest F1-score is selected as the optimal value.
This process allows the model to better balance precision and recall. In practical terms, it helps identify more churn customers, even if it slightly increases false positives.

5.6 Workflow Summary
The complete workflow of this study can be summarized as follows:

1. Data cleaning and preprocessing
2. Feature transformation and leakage removal
3. Training base models
4. Applying ensemble techniques
5. Evaluating model performance
6. Optimizing classification threshold

This step-by-step approach ensures that the models are not only accurate but also reliable and applicable in real-world scenarios.

6. Results and Analysis
This section presents a comprehensive analysis of the experimental results obtained from individual machine learning models, ensemble techniques, cross-validation performance, and

threshold optimization. The evaluation is based on multiple metrics including Accuracy, Precision, Recall, F1-score, and ROC-AUC, ensuring a balanced understanding of model behavior, especially under class imbalance conditions.

6.1 Performance of Base Models

Model	Accuracy	Precision	Recall	F1-score	ROC-AUC
Random Forest	0.8027	0.6778	0.4893	0.5683	0.8497
XGBoost	0.7857	0.6078	0.5428	0.5734	0.8294
MLP	0.7750	0.6432	0.3422	0.4468	0.8276
Logistic Regression	0.7161	0.4624	0.4278	0.4444	0.7117

Table-1: Performance of Base Models

The results of the base models show that **tree-based ensemble models (Random Forest and XGBoost)** significantly outperform simpler linear models such as Logistic Regression and neural approaches like MLP in this dataset.

Random Forest achieves the highest **accuracy (0.8027)** and strongest **ROC-AUC (0.8497)**, indicating that it is highly effective in separating churn and non-churn customers. However, its **recall (0.4893)** remains relatively low, suggesting that it still misses a considerable number of actual churn cases.

XGBoost demonstrates a more balanced performance, achieving the **highest recall (0.5428)** among all base models. This is particularly important in churn prediction, where identifying potential churners is more critical than overall accuracy.

The MLP model underperforms in recall (0.3422), indicating difficulty in capturing minority class patterns, likely due to dataset imbalance and limited feature scaling or tuning.

Logistic Regression serves as a baseline model and performs the weakest overall, confirming that the relationship between features and churn is highly nonlinear.

Overall, base model results highlight a key observation: **no single model simultaneously achieves high precision and recall**, motivating the use of ensemble methods.

6.2 Train vs Test Diagnostics (Overfitting Analysis)

Model	Train Accuracy	Test Accuracy	Accuracy Gap	Train F1	Test F1
XGBoost	0.9823	0.7857	0.1966	0.9665	0.5734
Random Forest	1.0000	0.8027	0.1973	1.0000	0.5683
MLP	0.8843	0.7750	0.1093	0.7341	0.4468
Logistic Regression	0.8981	0.7161	0.1820	0.8040	0.4444

Table-2: Overfitting Analysis

The train-test diagnostics reveal a clear presence of overfitting across most models, particularly in Random Forest and XGBoost.

Random Forest achieves a perfect training accuracy of 1.0000, but its test accuracy drops to 0.8027, resulting in a substantial gap (~0.197). This indicates that the model has learned patterns specific to the training data, including noise, rather than generalizable relationships.

Similarly, XGBoost shows strong training performance (0.9823 accuracy and 0.9665 F1-score) but suffers from a noticeable drop in test performance. This suggests that although boosting improves predictive power, it can still overfit when not properly regularized.

In contrast, MLP demonstrates a smaller accuracy gap (0.1093), indicating better generalization compared to tree-based models. However, its overall predictive capability is weaker, particularly in identifying churn cases.
Logistic Regression shows moderate overfitting but remains limited in performance due to its linear assumptions.
These results emphasize that high training performance does not necessarily translate to strong real-world performance**, reinforcing the need for ensemble regularization strategies and cross-validation.**

6.3 Performance of Ensemble Models

Model	Accuracy	Precision	Recall	F1-score	ROC-AUC
Bagging	0.8006	0.6555	0.5241	0.5825	0.8513
Stacking	0.8041	0.6713	0.5134	0.5818	0.8502
Voting (Hard)	0.7991	0.6553	0.5134	0.5757	—
Voting (Soft)	0.7842	0.6316	0.4492	0.5250	0.8293
AdaBoost	0.7346	0.0000	0.0000	0.0000	0.8268

Table-3: Ensemble Model Performance

Ensemble methods show a clear improvement in **stability and generalization** compared to individual models.
Among all ensemble techniques, **Stacking achieves the highest accuracy (0.8041)**, slightly outperforming Bagging. This indicates that combining multiple base learners through a meta-model helps capture complementary patterns in the data.
Bagging also performs strongly, with the **highest ROC-AUC (0.8513)** among all ensemble methods, suggesting excellent class separation ability.
Voting classifiers show moderate performance, with hard voting slightly outperforming soft voting in terms of recall and F1-score. However, soft voting suffers from reduced recall, indicating that probability averaging may dilute strong class signals in imbalanced settings.
A notable observation is the failure of AdaBoost, which produces **zero precision and recall**, indicating that the model is unable to correctly identify churn cases. This may be due to sensitivity to noisy or imbalanced data, where boosting misclassifies minority class patterns repeatedly.
Overall, ensemble methods especially **Bagging and Stacking demonstrate superior robustness**, balancing predictive performance and generalization ability.

6.4 Cross-Validation Performance

Model	CV Accuracy	CV F1-score	CV ROC-AUC
Bagging	0.8053	0.5829	0.8546
Stacking	0.8076	0.5818	0.8529
Voting (Hard)	0.8000	0.5600	—
Voting (Soft)	0.7897	0.5332	0.8282
AdaBoost	0.7346	0.0000	0.8371

Table-4: Cross Validation Performance

Cross-validation results confirm the reliability of ensemble models, particularly Bagging and Stacking, which consistently achieve stable performance across folds.

Stacking achieves the highest cross-validation accuracy (0.8076), closely followed by Bagging (0.8053). This consistency indicates that these models are not only accurate but also stable across different data splits.
Voting methods show slightly reduced stability, especially soft voting, which has lower F1-scores across folds. This suggests sensitivity to probability calibration.
AdaBoost again performs poorly, confirming its instability on this dataset.
Overall, cross-validation results reinforce that Bagging and Stacking are the most reliable ensemble approaches for churn prediction in this study.

6.5 Threshold Tuning Analysis

Setting	Accuracy	Precision	Recall	F1-score
Default (0.50)	0.8006	0.6555	0.5241	0.5825
Tuned (0.38)	0.7885	0.5826	0.7166	0.6427

Table-5: Threshold Tuning Analysis

Threshold optimization produces one of the most important improvements in this study.
At the default threshold of 0.50, the model is relatively balanced but biased toward higher precision and lower recall. This means fewer false positives but a significant number of missed churn cases.
After lowering the threshold to 0.38, recall increases dramatically from 0.5241 to 0.7166, meaning the model now captures significantly more churn customers. Although precision decreases slightly, this trade-off is acceptable in churn prediction, where identifying at-risk customers is more valuable than minimizing false alarms.
Most importantly, the F1-score improves from 0.5825 to 0.6427, indicating a better balance between precision and recall.
Verdict: Threshold tuning is as important as model selection in imbalanced classification problems.

6.6 Key Insights from Results

Across all experiments, several important findings emerge:

- Ensemble models consistently outperform individual models in stability and generalization.
- Tree-based models (Random Forest, XGBoost) perform strongly but tend to overfit.
- Stacking and Bagging provide the best balance of accuracy and ROC-AUC.
- AdaBoost is not suitable for this dataset due to instability under class imbalance.
- Threshold tuning significantly improves churn detection performance.
- Recall is more important than accuracy in churn prediction scenarios.

7. Discussion Conclusion

The results of this study provide a comprehensive understanding of how different machine learning and ensemble learning techniques perform in the context of customer churn prediction. Overall, the findings strongly indicate that ensemble learning methods outperform individual machine learning models, particularly in terms of stability, generalization, and balanced predictive performance.
From the evaluation of base models, it is evident that tree-based algorithms such as Random Forest and XGBoost outperform linear models like Logistic Regression and neural approaches like MLP. This suggests that churn behavior in the dataset is governed by complex, nonlinear relationships

among customer attributes such as tenure, contract type, service usage, and billing patterns. Although Random Forest achieves the highest accuracy and ROC-AUC among individual models, and XGBoost shows the best recall, both models still fail to simultaneously optimize precision and recall, which is crucial in churn prediction tasks.

The train-test diagnostic analysis further highlights a significant concern: overfitting in high-capacity models. Random Forest and XGBoost show large gaps between training and testing performance, indicating that these models tend to memorize training patterns rather than generalize effectively. On the other hand, simpler models such as Logistic Regression and MLP demonstrate smaller gaps but at the cost of reduced predictive power. This trade-off clearly shows that no single model provides an optimal solution, reinforcing the need for more robust learning strategies.

Ensemble learning techniques address many of these limitations by combining multiple models to improve overall performance. The results clearly show that Bagging and Stacking outperform other ensemble methods, achieving the most consistent and balanced results across accuracy, F1-score, and ROC-AUC metrics. Bagging, in particular, provides strong generalization and achieves the highest ROC-AUC, indicating excellent ability to distinguish between churn and non-churn customers. Stacking further improves predictive performance by learning an optimal combination of base learners through a meta-model.

In contrast, Voting classifiers show moderate performance, with hard voting performing slightly better than soft voting in terms of recall and F1-score. Soft voting, however, suffers from reduced performance, likely due to probability averaging weakening strong class-specific signals. A notable exception is AdaBoost, which performs poorly across all evaluation metrics, indicating that it is highly sensitive to class imbalance and noise in this dataset, making it unsuitable without further tuning or data preprocessing.

A particularly important finding of this study is the impact of threshold optimization. Adjusting the classification threshold from 0.50 to 0.38 significantly improves recall, increasing it from 0.5241 to 0.7166, and also improves the F1-score from 0.5825 to 0.6427. Although this adjustment slightly reduces precision and accuracy, the improvement in recall is highly valuable in churn prediction scenarios, where correctly identifying customers likely to churn is more important than minimizing false positives. This demonstrates that decision threshold tuning is a critical yet often overlooked step in building effective classification systems for imbalanced datasets.

Overall, this study demonstrates that ensemble learning combined with threshold tuning provides a highly effective framework for customer churn prediction. While individual models can capture certain patterns in the data, they are limited in their ability to generalize and handle class imbalance effectively. Ensemble methods, particularly Bagging and Stacking, offer a more robust and reliable solution by leveraging the strengths of multiple models.

In conclusion, the findings confirm that improving churn prediction performance is not solely dependent on selecting a powerful algorithm but also on applying appropriate ensemble strategies and optimizing decision thresholds. These insights are highly applicable in real-world business environments such as telecommunications, where early identification of potential churners can significantly enhance customer retention strategies and reduce revenue loss.

Future work can extend this study by incorporating advanced techniques such as SMOTE for handling class imbalance, exploring deep learning-based architectures, and investigating dynamic ensemble selection methods to further improve predictive accuracy and adaptability in evolving customer behavior patterns.